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Study on the influence of different fertilization regimes on the yield and bulb quality of garlic (Allium Sativum L.)

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Abstract

Purpose: The purpose of this study was to assess the effect of microbial biofertilizer and nanofertilizer on the growth, allicin content and elemental content of garlic in China.

Method: We conducted field trial of microbial biofertilizer and germanium-containing controlled release nanofertilizer along two consecutive winter periods of 2020/2021 and 2021/2022 in a major garlic production area at Pizhou city to determine the effect of treatment in comparison to the local fertilization practice.

Result: The results revealed that the application of microbial biofertilizer and germanium-containing controlled release nanofertilizer increased the bulb yield by approximately 6-28% and 14%, respectively, compared with local fertilization practice. Meanwhile, the application of microbial biofertilizer consistently increased the allicin content of garlic in year 2021 and 2022. Furthermore, we also observed a strong positive correlation between shoot nitrate concentration at spring garlic stage and the final garlic bulb yield, implying that the accumulation of nitrate content in garlic at spring garlic stage maybe beneficial to biomass accumulation through either nutritional effect and increased chilling tolerance.

Conclusion: Taken together, our results revealed that the application of microbial biofertilizer and germanium-containing controlled release nanofertilizer can increase growth and nitrogen use efficiency of garlic in our experiment.

1 Introduction

Garlic (*Allium sativum* L.) is one of the oldest known and popular bulb vegetable crops that is growing globally in temperate and tropical regions. Garlic belongs to the family Alliaceae, which also includes onion, chive and shallot (Takhtajan 1997). Garlic has often been used worldwide as flavoring for dishes attributed to its unique aroma. In addition, garlic has been proven to be the excellent nutritional source of manganese, vitamin C and vitamin B6 (Petropoulus et al. 2018). In traditional medicine, garlic is used to treat various illnesses such as hypertension, cancer, cold and coughing (Gebreyohannes et al. 2013). Besides that, garlic has been found to have been reported to have high antibacterial, antifungal, anti-protozoal and antiviral activities, indicating its potential in treating different diseases (Batiha et al. 2020). According to a latest population-based cohort study, anti-aging protective effect of garlic might be associated with a lower risk of all-cause mortality (Shi et al. 2019).

According to the FAO database, the total global production of garlic reached 30.7 million tons with a land area of 1.6 million hectares in 2019 (FAO 2019). Among garlic-producing countries, China is the world largest garlic producer with an annual output of 23.3 million tons and the production is in increasing trend to meet the demand of growing population. The total cultivated area for garlic in China has reached 0.83 million hectares (FAO 2019).

The growth, yield and quality attributes of garlic are strongly affected by environmental factors such as genotypes and environmental factors such as climatic conditions, soil properties and agronomic practices (Atif et al. 2019). Among the agronomic practices, the type and the rate of fertilizer application have a profound effect on the yield of garlic and on the sustainability of soil productivity (González et al. 2009). As a shallow rooted vegetable, garlic have relatively higher demand for fertilizer for optimal growth and development due to its lower nutrient extraction capacity. Lower nutrient extraction capacity and higher nutrient requirement of garlic will often lead to overapplication of fertilizer by farmers that subsequently caused environmental problems (Jiku et al. 2020). Hence, a better fertilizer practices that can reduce the application rate of fertilizer without compromising the yield of garlic is very much needed.

Conventional agricultural practices are highly dependent on chemical fertilizers and pesticides. While the chemical fertilizer can quickly increase crop yields, pro-longed use of chemical fertilizer can lead to harmful effect on the soil quality and environment (Savci 2012). Furthermore, the recent global energy crisis has also led to a tight supply of chemical fertilizer, and the price of chemical fertilizers has risen sharply, triggering food security concerns. Therefore, there is an urgent need for environmentally friendly agricultural practices that can improve fertilizer use efficiency. Microbial biofertilizer is one of the specific groups of fertilizer that has been shown to increase crop productivity through enhancing the availability and uptake of mineral nutrients for plants (Okur 2018). The underlying mechanisms by which microbial biofertilizer enhance plant growth are diverse, including biological nitrogen fixation, production of plant growth hormones, promoting solubilization of potassium and phosphorus (Kour et al. 2020). In garlic cultivation, studies reported that the combination of NPK fertilizer and microbial biofertilizer application can reduce fertilizer use without compromising the crop yield (Chanchan et al. 2018). Chanchan et al. (2018) showed that the production of garlic increased by about 15% after adding microbial biofertilizer, although the NPK doses was reduced by 25%. Hence, microbial biofertilizers have superior importance in nutrient management system to ensure agricultural sustainability, particularly for garlic cultivation.

Nanofertilizer is a new type of smart fertilizer developed by using nano-technology. Due to its controlled release properties and increased nutrient bioavailability, it has been well regarded as a novel technique towards attaining sustainable agriculture (Raliya et al. 2018). Nanofertilizer can be categorized into three different classes i.e., nanoscale supplements, nanoscale additives and fertilizers with nanoparticles coating (Dhir 2021). Among which, nutrient encapsulated/coated with nanomaterial coatings has been shown to have the effect of slowing down the release of nutrient and increase nutrient use efficiency of crops by minimizing nutrient leaching (Kottegoda et al. 2017). Besides that, the applica-tion of nanofertilizer can increase the bioavailability of nutrient due to its high specific surface area and high reactivity (Zulfiqar et al. 2019). Previous study showed that the application of “nano-urea” was beneficial to the growth of many crops including rice, radish, celery, cabbage, eggplant, pepper, tomato and so on. It was observed that the grain yield and N uptake of rice was significantly increased under “nano-urea” treatment (Sahu et al. 2022). However, the application of nanofertilizer in garlic cultivation is rarely reported. Therefore, in this study, we investigated the effect of microbial biofertilizer and nanofertilizer towards the yield, nutrient status and nutrient use efficiency of garlic

2. Materials And Method

2.1 Description of study site and experimental design

The garlic cultivar used in this study was Daqingke, a Jiangsu local guideline cultivar, which originated from Shandong province, China. The experiment was conducted in the experimental site located at Pizhou city (34º17'N, 117º41'E), Jiangsu province, China (Figure S1). The physicochemical properties of the soil in the first-year study were as follows: pH 7.63, 36.95 ± 11.06 g·kg^{-1} organic matter, 1.60 ± 0.22 g·kg-1 N, 96.57 ± 15.60 g·kg-1 P_2O_5 and 231.11 ± 48.50 mg·kg^{-1} K_2O. The size of each plot was 1000 m^2. The garlic was planted at 22 cm apart rows with 15 cm of individual spacing. The garlic was sown on 10 Oct 2020 and harvested on 14 May 2021. The variation in the ambient temperature throughout the growth period of garlic in this study was shown in Figure S2. In the second-year study, the physicochemical properties of the soil were as follows: pH 7.89, 16.82 ± 2.16 g·kg^{-1} organic matter, 0.85 ± 0.12 g·kg^{-1} N, 32.30 ± 6.98 g·kg^{-1} P_2O_5 and 274.03 ± 18.21 mg·kg^{-1} K_2O. The garlic was sown on 22 Oct 2021 and harvested on 20 May 2022. Table 1 showed the fertilization treatments used in the study. The first fertilizer used in this study was microbial biofertilizer (the product number in the market, Sunongfei2021lianzhuzhi003). The second fertilizer used in this study was germanium-containing controlled release nanofertilizer (the product number in the market, Sunongfei2021lianzhuzhi006). The preparation of germanium-containing controlled release nanofertilizer has been described in China (CN) Chinese patent documents with the publication number of CN108794114A.

Table 1
Treatment carried out on garlic (*Allium sativum* L.)

Treatment	Rate (kg·ha^{-1})	N (kg·ha^{-1})	P_2O_5 (kg·ha^{-1})	K_2O (kg·ha^{-1})	Other
40% 10-14-16 (Control)	1200	120	168	192	-
30% 8-10-12 (T1)	1500	120	150	180	Microbial Biofertilizer
30% 8-10-12 (T2)	1500	120	150	180	Germanium-containing Control Release Nanofertilizer

2.2 Biomass, total nitrogen (N) measurement and calculation of NUE

The garlic plant samples were divided into shoots and roots, and weighed to determine the fresh weight. The plant sample was then heated at 105°C for 30 min to inactivate cellular enzymatic activities. The total N concentration in garlic plant samples were determined using Kjeldahl method (Kjeldahl 1883).

Briefly, the finely ground garlic plant sample (3 g) were moistened with small amount of deionized water, then mixed with 5 ml of concentrated sulfuric acid and stirred overnight. The tubes containing sample solution were heated at 230°C for 30 minutes. The temperature was adjusted to 280°C and maintained for 30 minutes once smoke appeared in the tube. After that, H_2O_2 solution was added into the sample solution and shake gently until the solution become clear. The sample solution was then heated for another 10 minutes and the nitrogen concentration in the sample were analyzed using continuous Flow AutoAnalyzer III (AA3).

The total N accumulated in different parts of garlic plant samples were calculated by multiplying the N concentration with the corresponding biomass weight. The nitrogen use efficiency ($kg \cdot kg^{-1}$ N) was calculated as: NUE = Y/N, where Y is the yield of garlic ($kg.ha^{-1}$) and N is the total amount of N fertilizer applied ($kg \cdot ha^{-1}$); nitrogen absorption efficiency (NAE, $kg.kg^{-1}$) was calculated by total N content in whole plant ($kg.ha^{-1}$)/total amount of N fertilizer applied ($kg.ha^{-1}$); physiological N use efficiency (PNUE, $kg.kg^{-1}$) was calculated as bulb yield/total N accumulation at maturity, and nitrogen harvest index (NHI, %) was calculated as total N accumulation in garlic bulb at maturity/total N accumulation at maturity (Chen et al. 2016).

2.3 Plant nitrate content analysis

For the determination of nitrate content, plant samples (0.5 g) were homogenized in 5 mL distilled water and transferred to 10 mL centrifuge tubes. The samples were boiled for 30 minutes in 100°C water and allowed to cool down to room temperature. The sample solution was then centrifuged at 5000 ×g for 10 min, and the supernatant was transferred to a clean tube. The supernatant (0.1 mL) was then added with 0.4 mL of 5% sulfuric acid-salicylic acid solution, mixed thoroughly on a vortex mixer and kept at room temperature for 20 minutes. After that, 9.5 mL of sodium hydroxide solution (8%) was slowly added into the solution. Nitrate concentration was then measured using absorbance at 410 nm via spectrophotometer.

2.4 Elemental analysis of garlic

For the determination of other elements, plant samples were first ground into powder in liquid nitrogen. The finely ground plant sample (0.5 g) was then placed in the digestion tube and mixed overnight with 5 mL concentrated nitric acid. The tubes were placed on a digestion block and heated at 120°C until frothing stopped. Then, the temperature was raised to 180°C and an appropriate amount of $HClO_4$ was added until the solution become clear (Chen et al. 2016). The concentration of phosphorus, potassium, magnesium, calcium, zinc, boron, iron and sulfur in the garlic sample was then analyzed via ICP-OES (ScientificTM iCAP6300).

2.5 Allicin content analysis

For the determination of allicin, peeled fresh garlic (10 g) was mixed with 30 mL of methanol solution (methanol: water = 85:15) and mashed with tissue masher. Then, the sample solution was topped up with methanol solution to 50 ml. Then, the sample solution was centrifuged at 12,000 rpm at 4°C for 20 min. A

total of 1 mL of supernatant was added with 1.5 ml of methanol solution. Later, the solution was filtered using 0.45 μm pore size filter. The filtered solution (20 μL) was then subjected to HPLC analysis using YMC-Pack ODS-AC18 (25 cm x 4.6 mm) with a gradient program at a flow rate of 0.8 mL/min.

2.6 Statistical analysis

The research data were analyzed by one-way ANOVA using Duncan's multiple range tests ($P < 0.05$) to compare the means. Different letters indicated statistically significant differences at $p < 0.05$ among treatments. All statistical evaluations were performed by applying the IBM SPSS Statistics version 20 software (SPSS Inc., Chicago, IL).

3. Results

3.1 Effect of different fertilizer treatments on growth and yield of garlic

In this study, the fresh weight of garlic plants, garlic bulb and bulb yield per m^2 was determined to evaluate the effects of different fertilizer treatments on the growth of garlic under field condition (Fig. 1 and Table 2). In 2021, the average fresh weight of garlic plants under T1 treatment was the highest (200.7 ± 0.67 g), followed by T2 treatment (164.3 ± 2.73 g) and control treatment (136 ± 4.73 g). This corresponded well with the average fresh bulbs weight per plant. Garlic plants under T1 treatment recorded the highest average fresh bulbs weight per plant (74.3 ± 0.33 g), whereas the garlic plants under control treatment have the lowest average fresh bulbs weight per plant (58 ± 1.16 g). At harvest, the bulb yield per m^2 was the highest under T1 treatment (2.23 ± 0.01 kg), followed by T2 and control treatment (Table 2). A similar finding was observed in year 2022. For the average fresh weight of garlic plant, the highest was recorded under T2 treatment (182.0 ± 7.02 g), followed by T1 treatment (168.5 ± 4.19 g) and control treatment (146.3 ± 11.06 g). Meanwhile, the average fresh weight of garlic bulb was the highest under T1 treatment (97.3 ± 9.88 g), followed by T2 treatment (91.2 ± 5.66 g) and control treatment (69.2 ± 8.41 g). Similarly, the bulb yield per m^2 of the garlic grown under T2 and T1 treatment were approximately 14% and 6% higher than that of control, respectively (Table 2). These findings indicated that the addition of microbial biofertilizer and germanium-containing controlled release nanofertilizer had positive effects on the growth and yield of garlic even under the conditions of reduced P and K supply.

Furthermore, we showed that the allicin concentration in garlic increased significantly with the application of microbial biofertilizer. In 2021, the allicin concentration in the garlic under T1 treatment was the highest (0.355 g/kg), however there was no significant difference between control (0.243 g/kg) and T2 treatment (0.283 g/kg) (Fig. 2a). On the other hand, in 2022, both treatments significantly increased the allicin content with T1 treatment recorded the highest allicin concentration (0.65 g/kg) followed by T2 treatment (0.53 g/kg) (Fig. 2b).

Table 2 Fresh weight (FW) of garlic plant, garlic bulb and bulb yield per m^2 of garlic harvested in 2021 and 2022. Values are means ± SE (n≥6). Significant difference (P<0.05) is indicated by different lower case letters above bars as determined by Duncan's multiple range test.

Treatments	FW of garlic plant (g)		FW of garlic bulb (g)		Bulb yield per m^2 (kg)	
	2021	2022	2021	2022	2021	2022
Control	136.0±4.73c	146.3±11.06c	58±1.16c	69.2±8.41b	1.74±0.03c	2.43±0.11c
T1	200.7±0.67a	168.5±4.19b	74.3±0.33a	97.3±9.88a	2.23±0.01a	2.57±0.02b
T2	164.3±2.73b	182.0±7.02a	66.3±0.88b	91.2±5.66a	1.99±0.03b	2.77±0.03a

3.2 The nitrate concentration and content in garlic under different fertilization treatments

In this study, we observed that the fluctuation patterns of nitrate concentration in the shoot and root of garlic plants at different growth stages are consistent, regardless of the types of fertilizer treatment (Fig. 3). The concentration of nitrate in garlic leaf peaked at the spring garlic stage and began to decline sharply during the bulb development stage. However, we found that the shoot nitrate concentration at spring garlic stage increased significantly under T1 and T2 treatment. In 2021, T1 and T2 treatment increased the shoot nitrate concentration by 44% and 37%, respectively (Fig. 3a). Meanwhile, in 2022, the shoot nitrate concentration of garlic at spring garlic stage increased by 51–52% under the treatment of both biofertilizers (Fig. 3b).

In 2021, the garlic under T2 treatment had the highest root nitrate concentration, which was approximately 39% and 53% higher than that of control and T1 treatment, respectively (Fig. 3c). In 2022, T1 treatment recorded the highest root nitrate concentration at spring garlic stage, which was about 36.6% and 33.6% higher than that of T2 treatment and control treatment, respectively (Fig. 3d). However, there was no significant difference in nitrate concentration in the shoot and root of garlic plants under different treatments at harvest stage.

3.3 The nitrogen concentration and content of garlic under different fertilization treatments

In 2021, the shoot concentration of nitrogen in garlic peaked at the spring garlic stage and declined gradually from the bulb developmental stage to the harvesting stage (Fig. 4). The highest shoot concentration of nitrogen was recorded by control treatment (25.24 ± 0.75 g/kg) at spring garlic stage,

followed by T2 (15.54 ± 1.9 g/kg) and T1 treatment (11.41 ± 0.33 g/kg) (Fig. 4a). In 2022, the shoot concentration of nitrogen peaked at the seedling stage and reached its lowest value at harvest stage, irrespective of the treatments. The highest shoot concentration of nitrogen was recorded by the control treatment which was 23.55 ± 1.41 g/kg (Fig. 4a). Meanwhile, the maximum nitrogen content in the shoot was recorded at bulb developmental stage, irrespective of treatments and location (Fig. 4b). In 2021, we observed that the shoot nitrogen content in garlic plant under control treatment was the highest throughout different growth stages, except for harvest stage. At bulb developmental stage, the shoot nitrogen content of control garlic plant was 59% and 38% higher than that of T1 and T2 treatments, respectively. However, in 2022, the garlic under T1 treatment had the highest shoot nitrogen content (192.93 ± 0.99 mg) during bulb development stage (Fig. 4b).

For the root concentration of nitrogen, the garlic grown in 2021 had the highest root concentration of nitrogen in spring garlic stage. Among which, T2 treatment recorded the highest root concentration of nitrogen (27.29 ± 2.55 g/kg), followed by control (24.35 ± 2.48 g/kg) and T1 treatment (16.12 ± 2.24 g/kg) (Fig. 4c). Similarly, the garlic grown in 2022 had the highest concentration of nitrogen during the spring garlic stage. Among the treatment, the garlic grown under control treatment recorded the highest root concentration, which was 23.05 ± 0.33 g/kg (Fig. 4c). In 2021, the root nitrogen content of garlic under most treatments peaked at the bulb development stage. Among which, the highest root nitrogen content was recorded by the garlic under T2 treatment. In contrast, the root nitrogen contents of garlic planted in 2022 reached its peak at spring garlic stage and the highest root nitrogen content was recorded by control treatment (Fig. 4d).
In this study, the potassium content in the shoot and root of garlic plants was also quantitatively determined to explore the effect of different treatments on the potassium accumulation in garlic plants (Fig. 5). From the results, we observed that there is no significant difference between treatments except for the root potassium concentration. For the two-years study, the root potassium concentration of garlic under treatment T1 and T2 consistently higher than that of control treatment. In 2021, the root potassium concentration of garlic under treatment T1 and T2 was approximately 41% and 32% higher than those of control treatment, respectively (Fig. 5c). Similarly, in 2022, the root potassium concentration of garlic under treatment T1 and T2 was approximately 30% and 19% higher than those of control treatment, respectively (Fig. 5c).

3.5 Elemental analysis of garlic bulb

Plant Mg, Ca, Fe, B and Zn content were measured to investigate the effect of different fertilization treatments on these nutrients in garlic bulb at harvest stage (Table 3). In 2021, it was observed that the Fe content in garlic bulb under T1 and T2 treatment was significantly higher than that in the control. The accumulation of Fe in garlic bulb under T1 and T2 treatment was approximately 63% and 64% higher than that of control treatment, respectively. Meanwhile, there was no significant difference in Mg, Ca, B and Zn content among different treatments.

In 2022, the Ca content in garlic bulbs under T1 treatment was significantly higher than that in the control and T2 treatment. Contrary to the data from year 2021, the Fe content in the garlic bulb under T1 and T2

treatment was significantly lower than that in the control. The accumulation of Fe in garlic under T1 and T2 treatment was approximately 17% and 23% lower than that in the control, respectively.

Table 3
Elemental analysis of garlic bulb cultivated under different fertilization treatments. Values are means ± SE (n ≥ 3). Significant difference (P < 0.05) is indicated by different lower-case letters as determined by Duncan's multiple range test.

Year	Element ($mg.kg^{-1}$)	Control	T1	T2
2021	Mg	3698.90 ± 227.83a	4111.22 ± 152.45a	4248.40 ± 322.40a
	Ca	2422.80 ± 114.36b	2573.72 ± 57.07ab	2630.87 ± 131.75ab
	Fe	190.12 ± 3.41b	310.94 ± 16.83a	312.00 ± 7.41a
	B	11.61 ± 0.49a	12.62 ± 0.69a	13.57 ± 1.36a
	Zn	11.94 ± 0.58a	11.14 ± 1.36a	10.92 ± 1.65a
2022	Mg	4130.73 ± 120.81a	3865.33 ± 139.59a	3887.10 ± 125.94a
	Ca	2382.15 ± 110.72b	2737.28 ± 203.61a	2448.31 ± 86.43ab
	Fe	196.14 ± 4.50a	161.93 ± 3.92b	150.22 ± 2.95c
	B	12.82 ± 0.63a	13.64 ± 1.13a	12.89 ± 1.27a
	Zn	9.90 ± 1.32a	8.89 ± 0.71a	9.26 ± 0.45a

Note: Mg: Magnesium; Ca: Calcium; Fe: Ferum; B: Boron; Zn: Zinc.

3.6 Nitrogen use efficiency (NUE) in garlic under different fertilization treatments

In view of the increase in biomass and bulb fresh weight of garlic grown under T1 and T2 treatment, we calculated the nitrogen use efficiency (NUE), nitrogen absorption efficiency (NAE), physiological N use efficiency (PNUE) and nitrogen harvest index (NHI) of garlic at the harvesting stage (Table 4). In 2021, the NUE of garlic under T1 and T2 treatments increased by about 28% and 14%, respectively. Likewise, the PNUE of garlic grown under T1 and T2 treatment increased by 46% and 112%, respectively. For NHI, the garlic under T2 treatment increased by about 60%. In contrast, the NAE of the garlic under T1 and T2 treatment decreased by approximately 12% and 46%. However, the NAE and NHI of T2 treatment decreased by about 46% and 15%, respectively compared to those of control. Similarly, in 2022, the NUE of the garlic under T2 treatment increased by about 5% relative to that of control treatment. Meanwhile, the NAE of the garlic under T1 and T2 treatment increased by 46% and 92%, respectively. In contrast, the PNUE of garlic under T1 and T2 treatment decreased by about 21% and 22%, respectively. There was no significant difference in NHI among different treatment.

Table 4
Comparison of nitrogen use efficiency, nitrogen absorption efficiency, physiological N-use efficiency and nitrogen harvest index between garlic grown under different fertilization treatments. Values are means ± SE (n ≥ 3). Significant difference (P < 0.05) is indicated by different lower-case letters as determined by Duncan's multiple range test.

	Parameter	Control	T1	T2
2021	NUE ($kg.kg^{-1}$)	145.0 ± 2.89c	185.8 ± 0.83b	165.8 ± 2.20a
	NAE ($kg.kg^{-1}$)	1.17 ± 0.05a	1.03 ± 0.02b	0.63 ± 0.01b
	PNUE ($kg.kg^{-1}$)	124.39 ± 5.10c	181.24 ± 3.97b	263.75 ± 1.70a
	NHI (%)	48.0 ± 0.10a	47.0 ± 0.10b	76.8 ± 0.4c
2022	NUE ($kg.kg^{-1}$)	201.8 ± 19.56a	203.5 ± 21.17a	210.94 ± 12.81a
	NAE ($kg.kg^{-1}$)	1.06 ± 0.53a	1.55 ± 0.21a	2.04 ± 0.15b
	PNUE ($kg.kg^{-1}$)	134.39 ± 4.65a	132.40 ± 5.34a	105.23 ± 13.87a
	NHI (%)	46.0 ± 0.40b	46.0 ± 0.10b	47.0 ± 0.20a

Note: NUE: Nitrogen Use Efficiency; NAE: Nitrogen Absorption Efficiency; PNUE: Physiological N-use efficiency; NHI: Nitrogen Harvest Index.

3.7 Correlation between nitrate concentration at spring garlic stage and garlic biomass

In this study, Pearson correlation analysis was conducted to investigate the relationship between nitrate concentration at spring garlic stage and the biomass of garlic at harvesting stage (Fig. 6). We observed that nitrate concentration in the shoot of garlic at spring garlic stage was positively correlated with to the biomass of garlic at harvest stage (R2 = 0.797; $p \leq 0.01$). Meanwhile, there was no correlation observed between nitrate concentration in the root of garlic and biomass of garlic at harvesting stage. The finding indicated that the shoot nitrate concentration at spring garlic stage was somewhat related to the final biomass of garlic.

4. Discussion

4.1 Nanofertilizer improve the biomass accumulation and quality parameter of garlic

In this two-years study, we showed that the application of microbial biofertilizer and germanium containing controlled release nanofertilizer significantly increased garlic plant biomass accumulation and bulb yield in a major garlic production area (Pizhou city) in China (Fig. 1; Table 2). In addition, the NUE and NAE of the garlic under the treatment of both fertilizers was also higher than those of control

treatment. The yield increasing effect of beneficial microorganisms on garlic has also been reported in several previous study. For example, the introduction of genetically engineered strain of the nitrogen-fixing bacterium *Pseudomonas protegens* CHA0 has been found to increase the bulb yield of garlic by 12.03% (Wang et al. 2020). Besides that, the study also revealed that the introduction of *Pseudomonas protegens* can lower disease index and increased the allicin content in garlic (Wang et al. 2020). This was consistent with our result that the allicin content in garlic was significantly increased under treatment of microbial biofertilizer. For nanofertilizer, a similar observation was also made by many previous studies that the application of CuO nanoparticle and Ag nanoparticle can significantly improve the agronomic traits of onion such as biomass accumulation, nutrient elements and yield (Fouda et al. 2020; Wang et al. 2020).

4.2 Pattern of nitrate accumulation throughout growth stage

In aerobic agricultural soil, nitrate is arguably the main source of mineral nitrogen for plants (Dechorgnat et al. 2011). Since garlic performed well when planted in well-drained soil, hence it is reasonable to assume that nitrate is essential for its growth and development. In plant, nitrate is actively taken up by root cells and mobilized to other plant organs through the help of nitrate transporters (Fan et al. 2017). Within plant cells, some of the absorbed nitrate is metabolized directly into nitrogen-containing compounds and the excessive nitrate is translocated into vacuole for storage (Wang et al. 2014). It was shown that cold temperature can have multiple negative effects on plant morphology and physiology, especially on root metabolic activity, leaf water potential and nitrate uptake (Laine et al. 1994; Burchett et al. 2006; Sun et al. 2017). In this study, we observed that the nitrate accumulation in the whole plant of garlic increased drastically during spring garlic stage. This nitrate accumulation in plants may be associated to the low temperature and then low photosynthesis and C accumulation during winter season. However, high nitrate from N accumulation at the later stage would result into a faster conversion rate of nitrate into N-containing compounds (such as protein) and therefore a faster biomass accumulation with high photosynthesis together at the late growth stage. However, at the later growth stage (i.e., bulb development stage and harvest stage), the conversion of nitrate into N-containing compounds become faster than the nitrate absorbed from the root, leading to lower nitrate accumulation.

4.3 Nitrate concentration at spring garlic stage was positively correlated with garlic biomass at harvest

In this study, we observed a strong positive correlation between the shoot nitrate concentration at spring garlic stage and garlic biomass. In contrast, there was no significant correlation between the root nitrate concentration and garlic biomass. This observation suggested that the shoot nitrate concentration in garlic at spring garlic stage exert significant impact on biomass accumulation of garlic at harvest. The observed phenomenon may be either due to the role of nitrate as nutrient or as the major anionic osmotica that increase plant tolerance to chilling stress (Heerden et al. 2004). Chilling stress is one of the environmental factors that restrict the growth and productivity of plant. Generally, chilling stress causes a decrease in the hydraulic conductivity of the roots and subsequently a significant decrease in leaf water potential. The reduced leaf water potential leads to the loss of leaf turgor pressure, resulting in severe

plant wilting (Hussain et al. 2018). Hence, osmotic adjustment is crucial to assists the plant to maintain higher turgor potential. It has been well established that nitrate is the major osmolyte that help maintaining the cell turgor by reducing the osmotic potential of vacuolar sap in many crops (Mott and Steward 1972). Besides that, previous study reported a linear correlation between plant nitrate and water content (Cárdenas-Navarro et al. 1999). For example, nitrate has been shown to involve in stomatal functioning by controlling the turgor pressure of guard cell (Guo et al. 2003). In addition, study showed that intracellular nitrate can also control the root hydraulic properties in plants, resulting into higher water uptake (Gorska et al. 2008). A previous study also showed that nitrate supplementation can significantly reduce the detrimental effect of chilling stress on crops (Heerden et al. 2004). This observation suggested that the higher shoot nitrate concentration at spring garlic stage can increase the chilling tolerance of garlic, which may be achieved by maintaining leaf water potential. In addition to nitrate, studies showed that potassium (K^+) is also important for cold tolerance in plant by regulating the osmotic and water potential, increasing antioxidant levels to reduce ROS production and reducing electrolyte leakage caused by chilling stress (Oosterhuis et al. 2013; Hasanuzzaman et al. 2018). Hence, the relatively higher root concentration of K^+ maybe a protective strategy for garlic to survive under chilling stress (Fig. 5c). As the major cation in plant cell, K^+ has been shown to be closely linked with NO_3^- (counterion of K^+) in translocation, transportation and distribution due to the effect of anion-cation balance in plant cells (Raddatz et al. 2020). Hence, it is possible that nitrate and potassium act synergistically in protecting the garlic from chilling stress. It has been reported that exogenous application of potassium nitrate can promote the cold tolerance in Panax ginseng by enhancing the antioxidant level (Devi et al. 2012). However, a more detailed study at the molecular level should be conducted in order to further understand the underlying mechanism.

5. Conclusions

Overall, the microbial biofertilizer and germanium-containing controlled release fertilizer used in this study could significantly increase the growth, nitrogen use efficiency and allicin content. This study also observed a strong positive correlation between the shoot nitrate concentration at spring garlic stage and garlic biomass. It indicated that higher shoot nitrate at spring garlic stage may promote the growth of garlic at later stage, which may be due to the synergistic effect of nitrate and potassium in enhancing the cold tolerance of garlic. Our study provides a scientific theoretical basis for high yield garlic using the microbial biofertilizer.

Declarations

Ethics approval and consent to participate: Here is no ethics conflicts in these data

Consent to publish: we are consent to publish these data

Availability of data and materials: The data and materials in the manuscript are available.

Competing interests: The authors declare no competing interests.

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Authors' Contributions: ZY produced the fertilizer; ZY, FX, ZX designed the experiment; SJ, LZ, FX conducted and wrote the manuscript; ZH and ZL supplied the experiment land.

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Figures

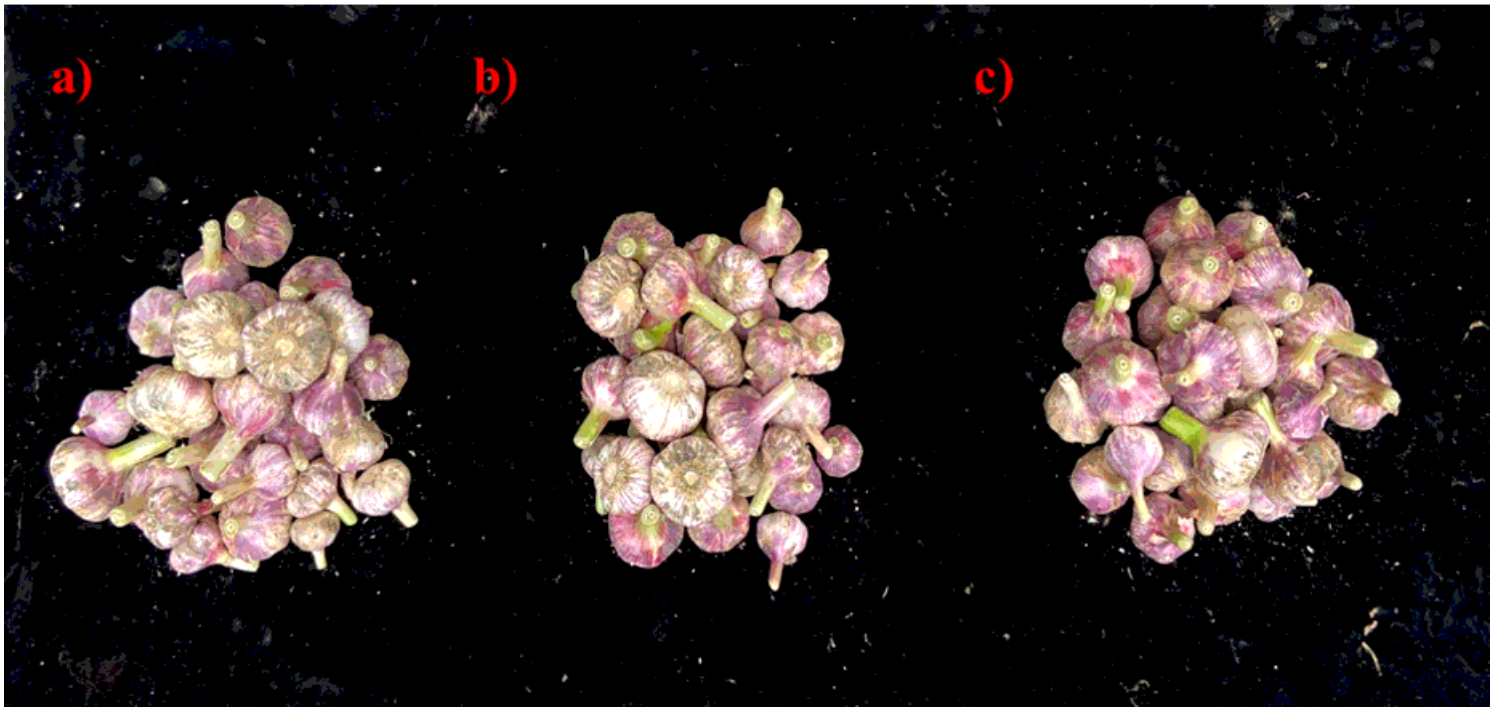

Figure 1

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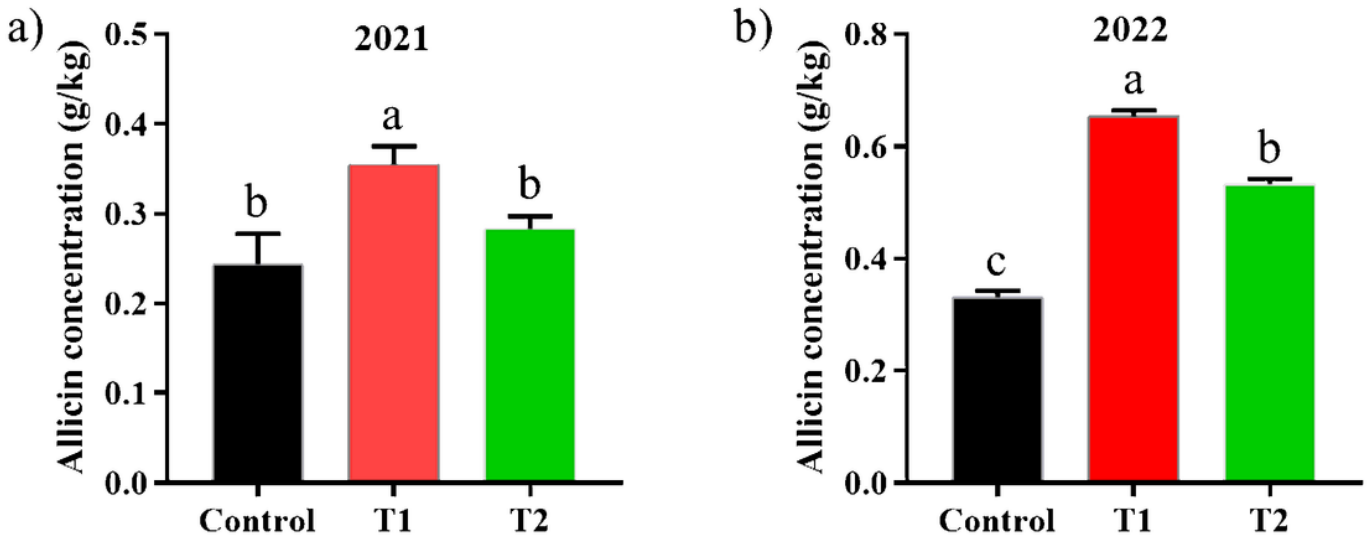

Figure 2

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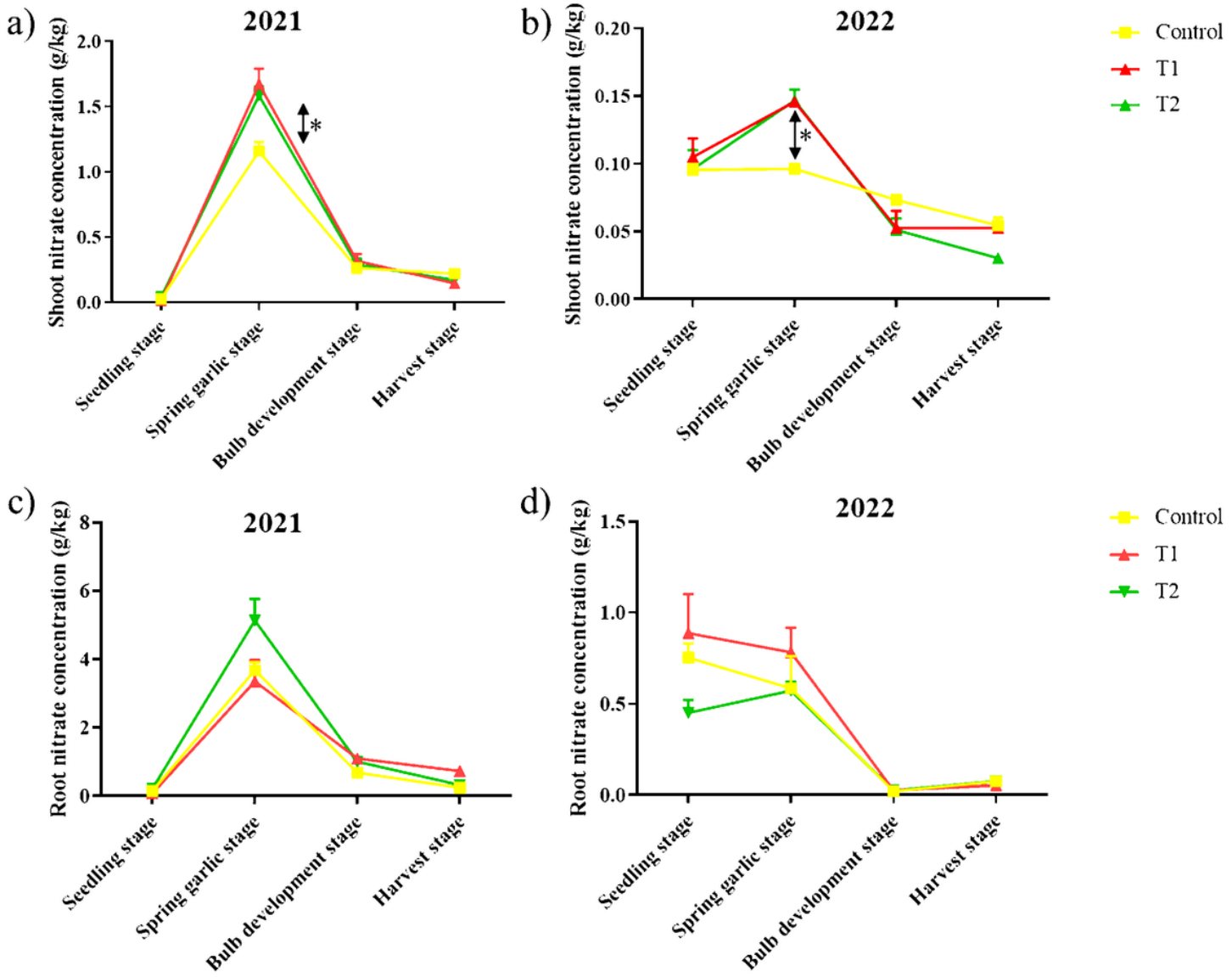

Figure 3

Fluctuation of nitrate concentration and content in the shoot and root of garlic plant at different growth stages under varying fertilizer treatments. **a** Shoot nitrate concentration ($g \cdot kg^{-1}$) of garlic harvested in 2021, **b** Shoot nitrate concentration ($g \cdot kg^{-1}$) of garlic harvested in 2022, **c** Root nitrate concentration ($g \cdot kg^{-1}$) of garlic harvested in 2021, **d** Root nitrate concentration ($g \cdot kg^{-1}$) of garlic harvested in 2022. Values are means ± SE ($n \geq 3$). Significant difference ($P<0.05$) is indicated by asterisk as determined by Duncan's multiple range test

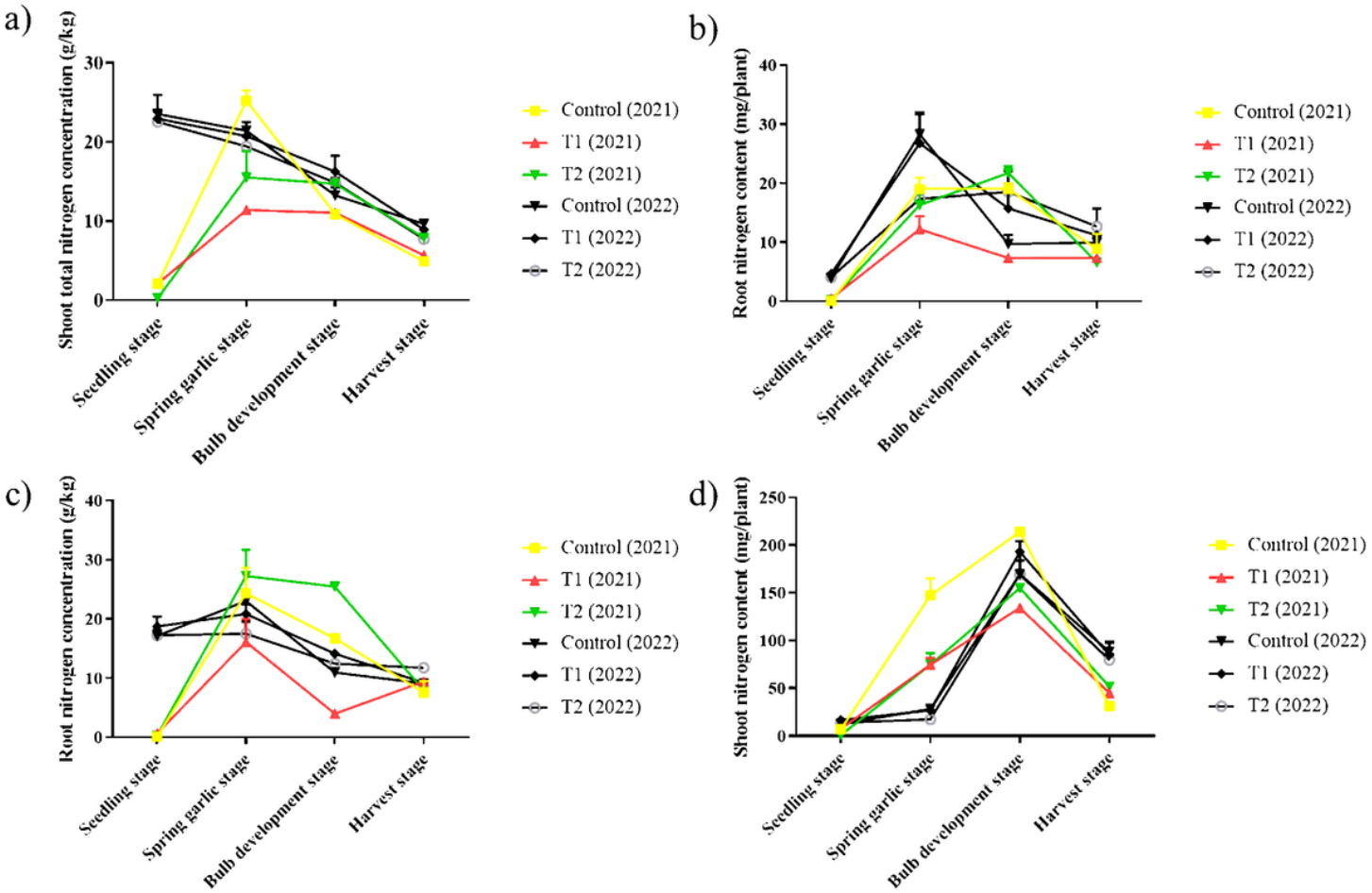

Figure 4

Nitrogen concentration and content in the shoot and root of garlic plant under different fertilizer treatments. **a** Nitrogen concentration in the shoot of garlic plant. **b**Nitrogen content in the shoot of garlic plant. **c** Nitrogen concentration in the root of garlic plant. **d** Nitrogen content in the root of garlic plant. Values are means±SE ($n \geq 3$). Significant difference ($P<0.05$) is indicated by asterisk as determined by Duncan's multiple range test

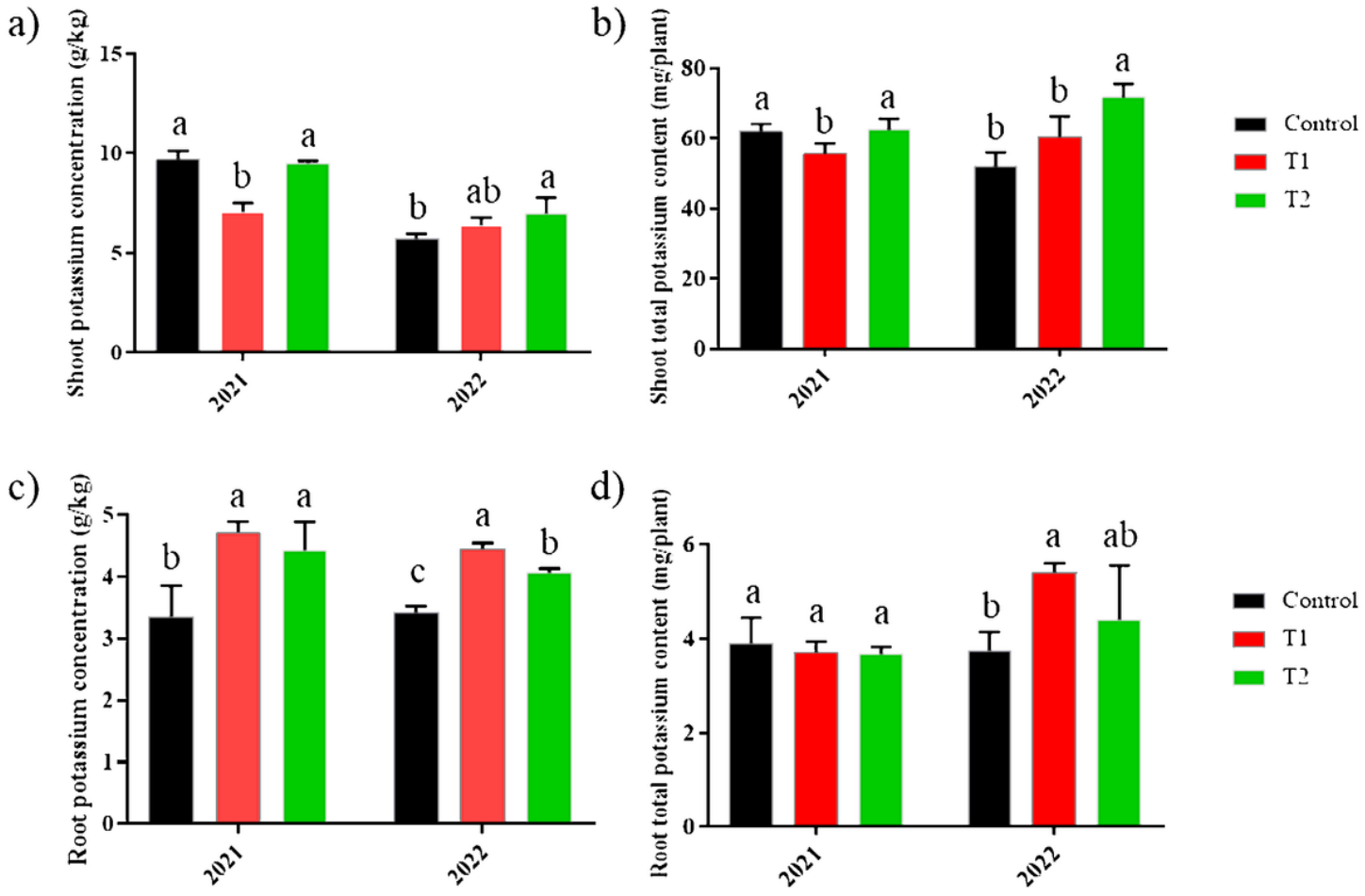

Figure 5

Available potassium concentration and content in the shoot and root of garlic plant under different fertilizer treatments. **a** Available potassium concentration in the shoot of garlic plant. **b** Available potassium content in the shoot of garlic plant. **c** Available potassium concentration in the root of garlic plant. **d** Available potassium content in the root of garlic plant. Values are means ± SE ($n \geq 3$). Significant difference ($P<0.05$) is indicated by asterisk as determined by Duncan's multiple range test

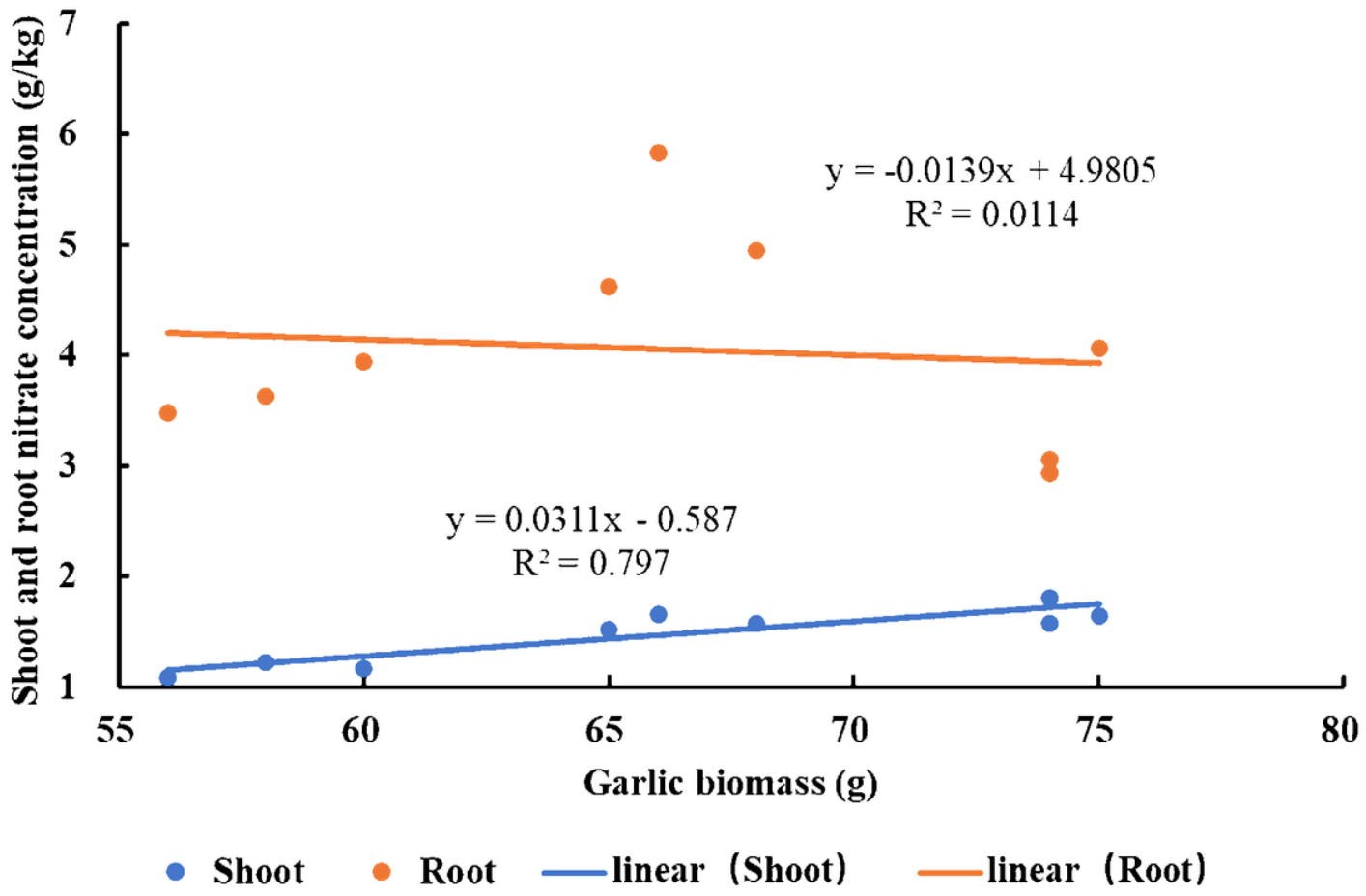

Figure 6

Pearson correlation analysis between nitrate concentration at the spring garlic stage and garlic biomass at harvest. Blue colored line represents pearson correlation analysis between shoot nitrate concentration at spring garlic stage and garlic biomass at harvest. Orange colored line represents pearson correlation analysis between root nitrate concentration at spring garlic stage and garlic biomass at harvest.

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